

CONVOLUTIONAL CODES

Simulation of Convolutional Codes and Analysis of Encoded Output Bit Sequences Using MATLAB

AIM : To generate and study Convolutional Codes and Analysis of Encoded Output Bit Sequences Using MATLAB .

SOFTWARE REQUIRED : MATLAB

Objectives

(i) To implement convolutional encoding for a given binary input sequence using a rate $\frac{1}{2}$ convolutional encoder and tabulate the corresponding encoded output pairs.

```
CODE : clc;

clear;

close all;

% Fixed input sequence
input_bits = [1 0 0 1 1];

% Convolutional encoder
trellis = poly2trellis(3,[7 5]);

encoded_bits = convenc(input_bits,trellis);

% Separate encoded bits into pairs
encoded_pairs = reshape(encoded_bits,2,[]);

% Create output strings
for i = 1:length(input_bits)
    output_pair{i} = sprintf('%d%d', ...
        encoded_pairs(i,1),encoded_pairs(i,2));
end

% Create table
Bit_Position = (1:length(input_bits));
Input_Bit = input_bits';
Output_Encoded = output_pair';

T = table(Bit_Position,Input_Bit,Output_Encoded);

disp('Convolutional Encoding Table');
```

```

disp(T);

% Plot

figure;

subplot(2,1,1)

stairs(input_bits,'LineWidth',1.5)

ylim([-0.2 1.2])

title('Input Bit Sequence')

xlabel('Bit Position')

ylabel('Bit Value')

grid on

subplot(2,1,2)

stairs(encoded_bits,'LineWidth',1.5)

ylim([-0.2 1.2])

title('Encoded Output Bit Sequence')

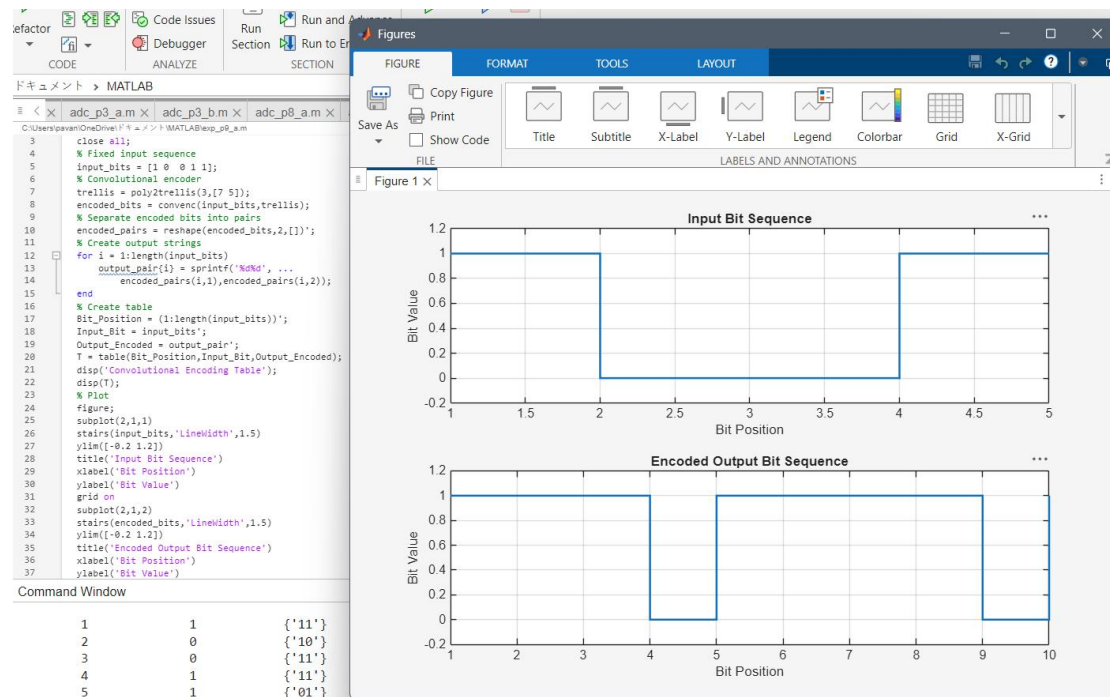
xlabel('Bit Position')

ylabel('Bit Value')

grid on

```

OUTPUT :



RESULTS : This program is successfully executed using the matlab .